ok i'll try to make as much sense as i can. ooh also first time using latex formally informally yay

1 theory

kepler's third law states that the square of the orbital period of a planet (P) is directly proportional to the cube of the semi-major axis of its orbit (a). this means that if you know the distance of a planet from the sun, you can calculate how long it takes to complete one orbit around the sun.

mathematically,

$$P^2 \propto a^3 \tag{1}$$

when we compare two planets orbiting the same star/central body, it forms a ratio,

$$\frac{P_1^2}{a_1^3} = \frac{P_2^2}{a_2^3} \tag{2}$$

if we measure the distances in AU, and the orbital periods in years, the constant of proportionality becomes 1 for all planets in our solar system,

$$P^2 = a^3 \tag{3}$$

kepler developed this law empirically, newton then used his law of universal gravitation and refined it,

for a circular orbit, the gravitational force is equal to the centripetal force,

$$\frac{GMm}{r^2} = \frac{mv^2}{r} \tag{4}$$

m cancels out, and we can substitute v with $\frac{2\pi r}{P}$,

$$\frac{GM}{r^2} = \frac{4\pi^2 r}{P^2} \tag{5}$$

which gives us kepler's third law in terms of the gravitational constant and mass of two bodies:

$$P^2 = \frac{4\pi^2}{G(M+m)} a^3 \tag{6}$$

c'est tout, je crois. la reste, c'est backward solving et je pense ça doit travailler- grace a dieu